Status and future of nonadiabatic dynamics with similarity-constrained coupled cluster theory

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Why coupled cluster theory, and why not?

Why yes?

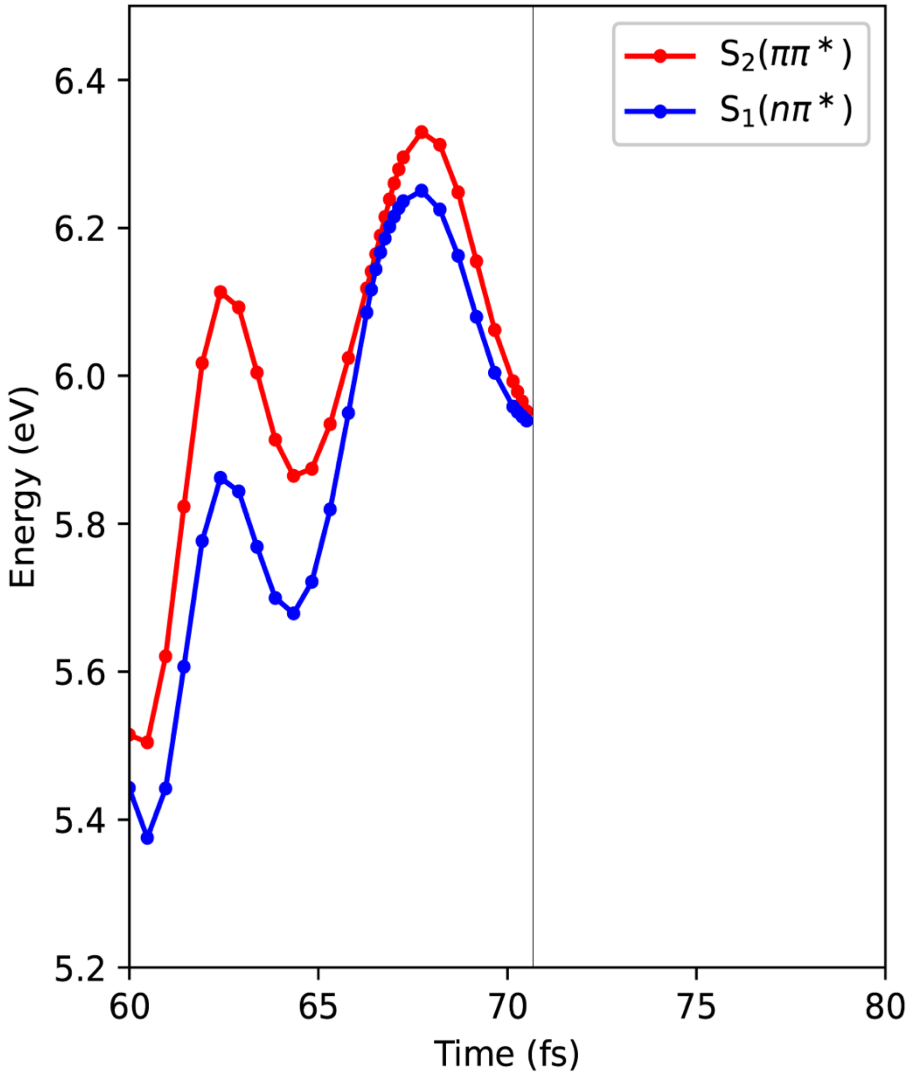
- High accuracy-to-cost ratio
- Systematically improvable (CC2, CCSD, CC3, …)
- Black-box

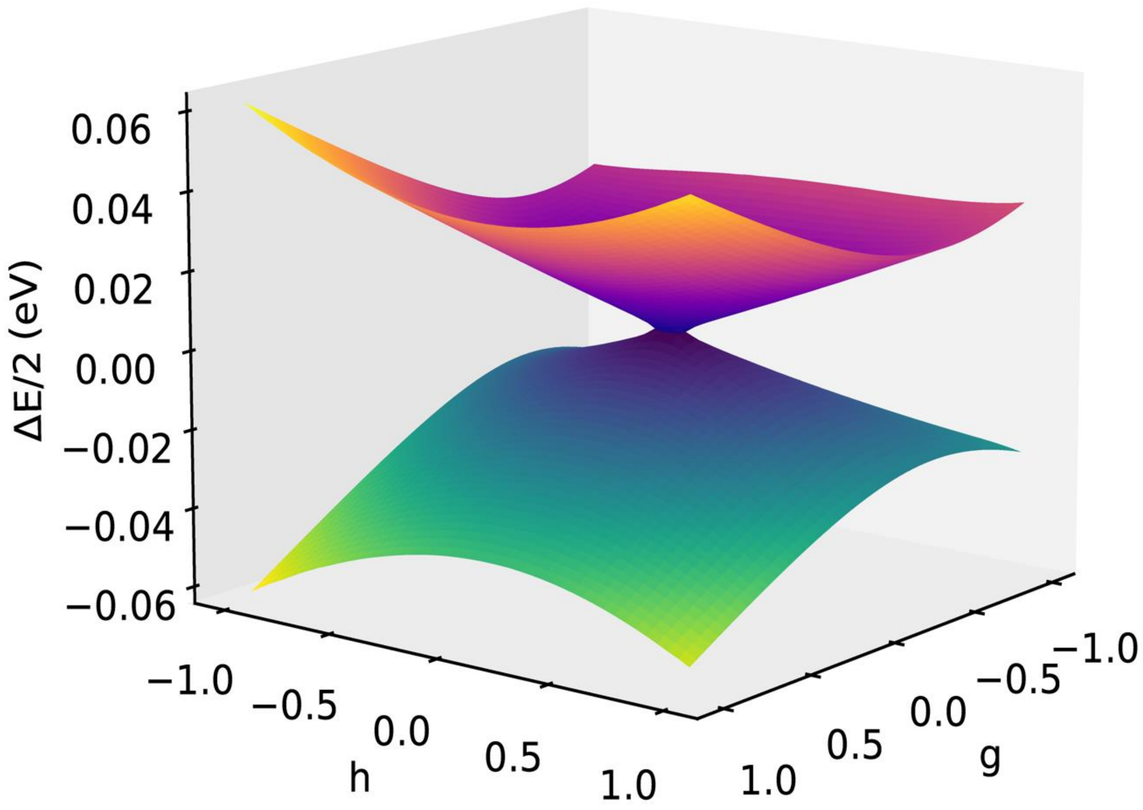
Why not?

- **Single-reference**: bias towards reference; S_0/S_1 not possible
- **Non-Hermitian:** numerical problems at conical intersections; S_1/S_2 *etc.* not possible

The problem…

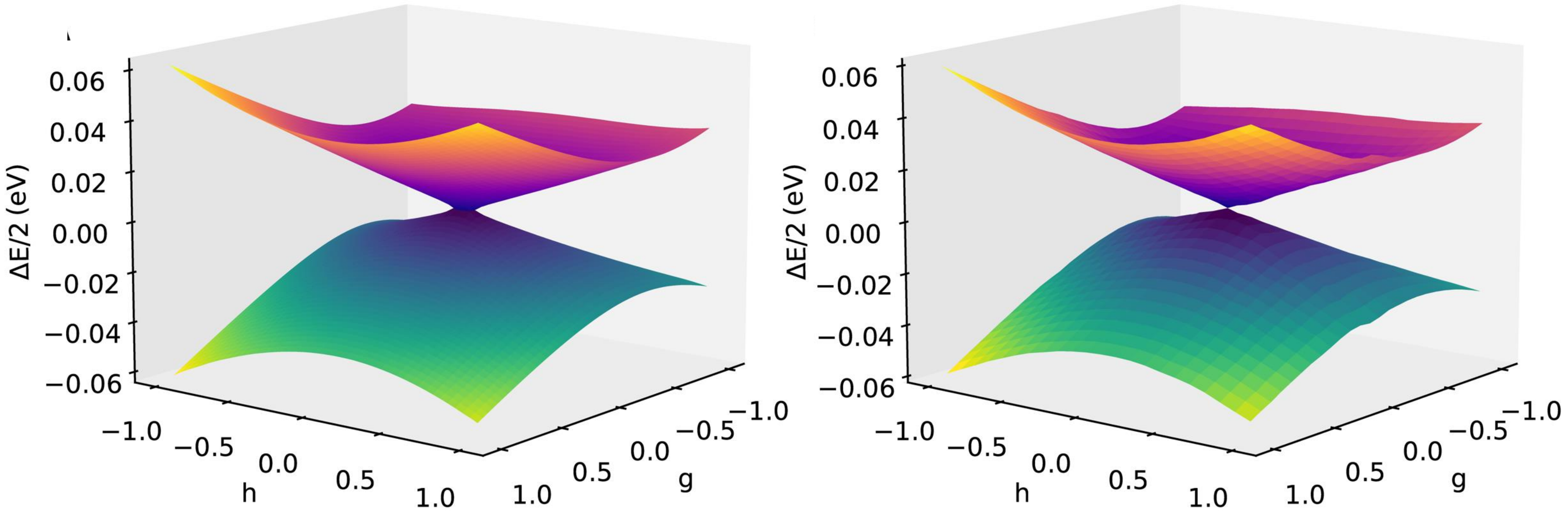
Köhn, Andreas, and Attila Tajti. *J. Chem. Phys* 127.4 (2007)
Plasser et al., *J. Chem. Theory Comput.,* 10, 4, 1395–1405 (2014)





Similarity-constrained coupled cluster theory[1]

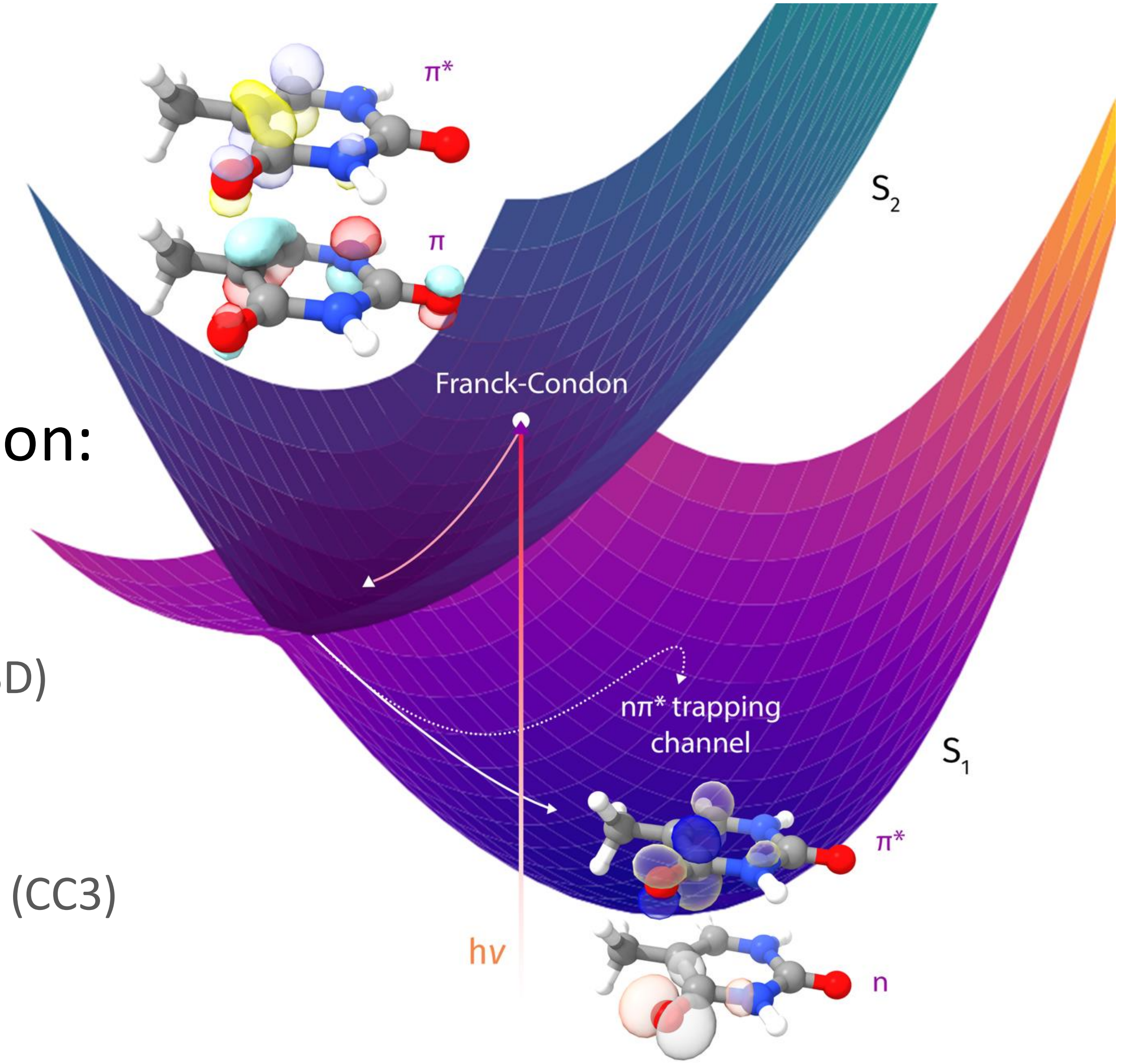
- Enforce similarity to a Hermitian theory by adjusting basis



1. Kjønstad and Koch, J. Phys. Chem. Lett. 2017, 8, 19, 4801–4807

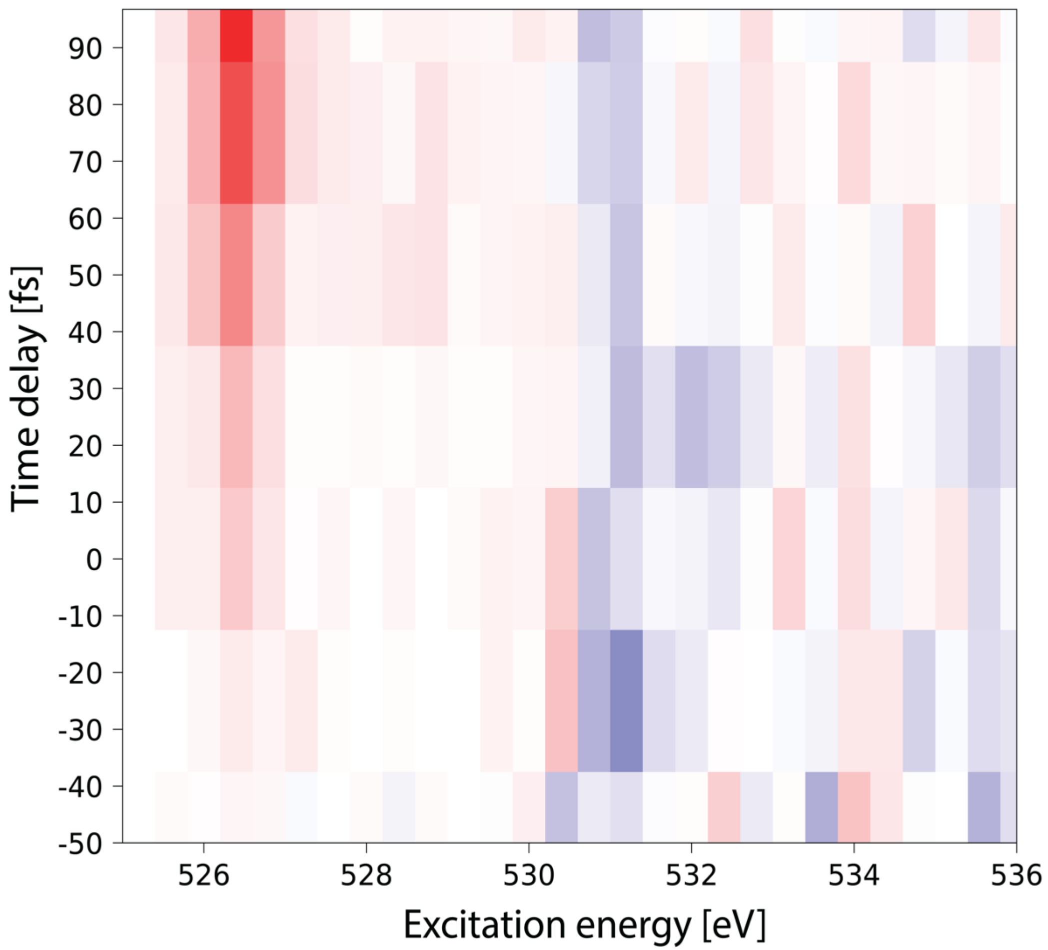
Thymine[2,3]

- Ultrafast (< 100 fs) ππ* to *n*π* relaxation
- Simulation-vs-experiment comparison:
  - Dynamics:
    - *Ab initio* multiple spawning (AIMS)
    - Coupled cluster singles and doubles (CCSD) with similarity constraints
  - X-ray absorption spectra:
    - Coupled cluster with perturbative triples (CC3)

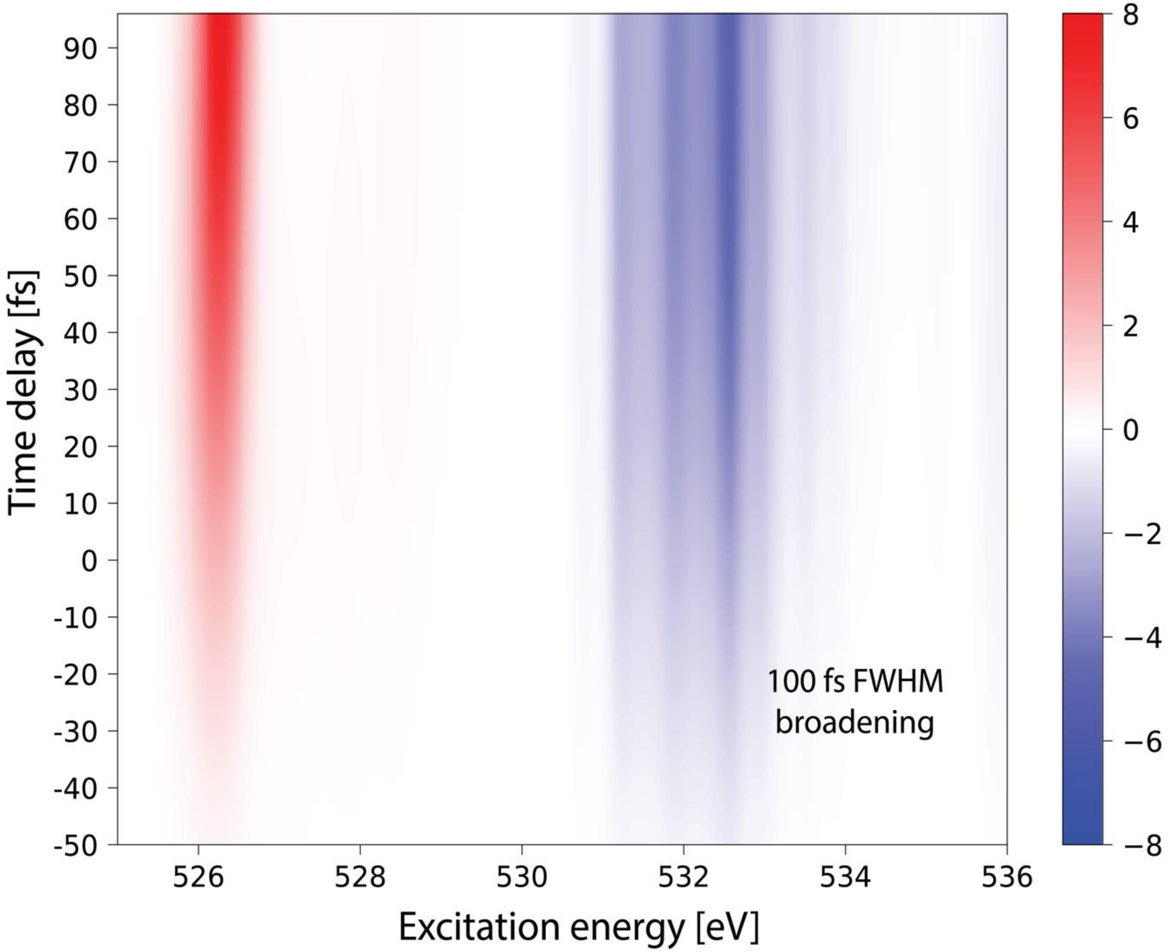


2. T. J. A. Wolf et al., Nature Communications 8, 29 (2017)
3. E. F. Kjønstad et al., Nature Communications, 15, 10128 (2024)

Thymine



Time delay [fs]
90
80
70
60
50
40
30
20
10
0
-10
-20
-30
-40
-50
526
528
530
532
534
536
Excitation energy [eV]



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0
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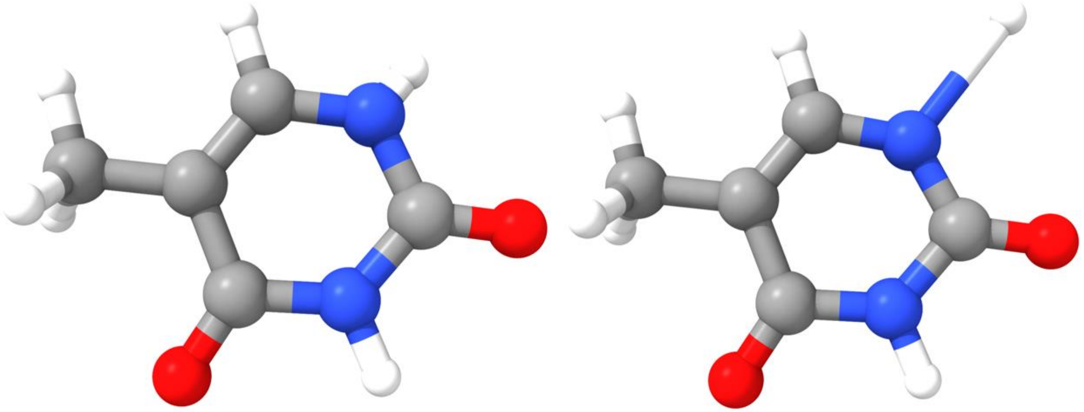
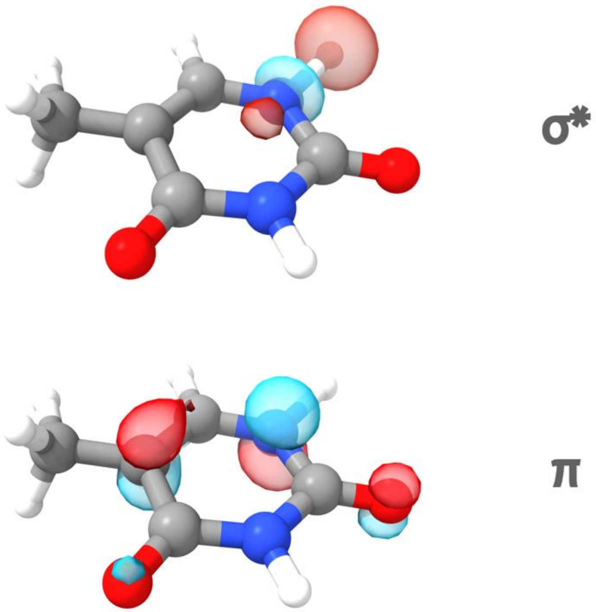
-20
-30
-40
-50
526
528
530
532
534
536
Excitation energy [eV]
100 fs FWHM
broadening
8
6
4
2
0
−2
−4
−6
−8

A $\pi\sigma^*$ NH-dissociation pathway?

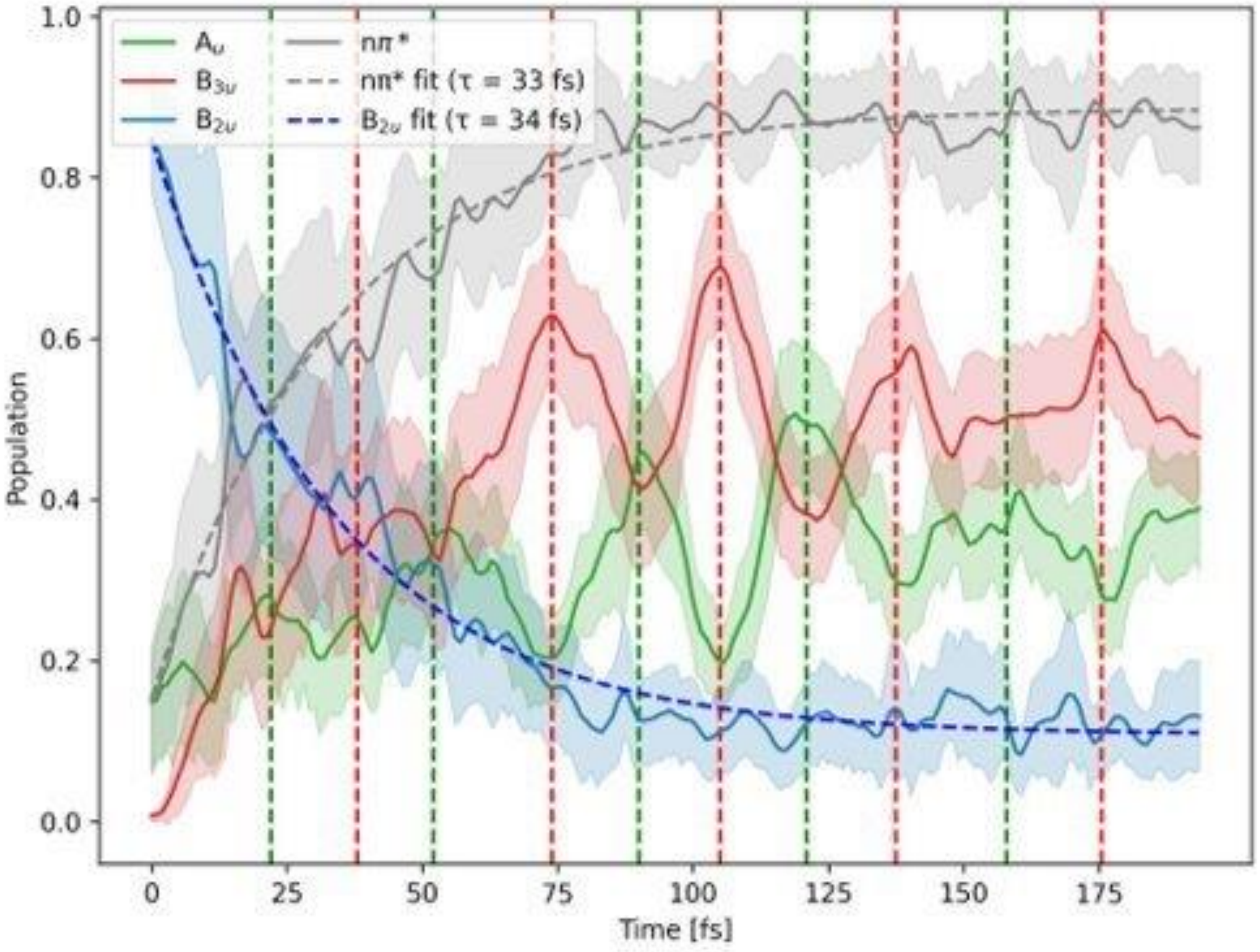




Similar pathways exist in other systems
(e.g., adenine)

Pyrazine[4]

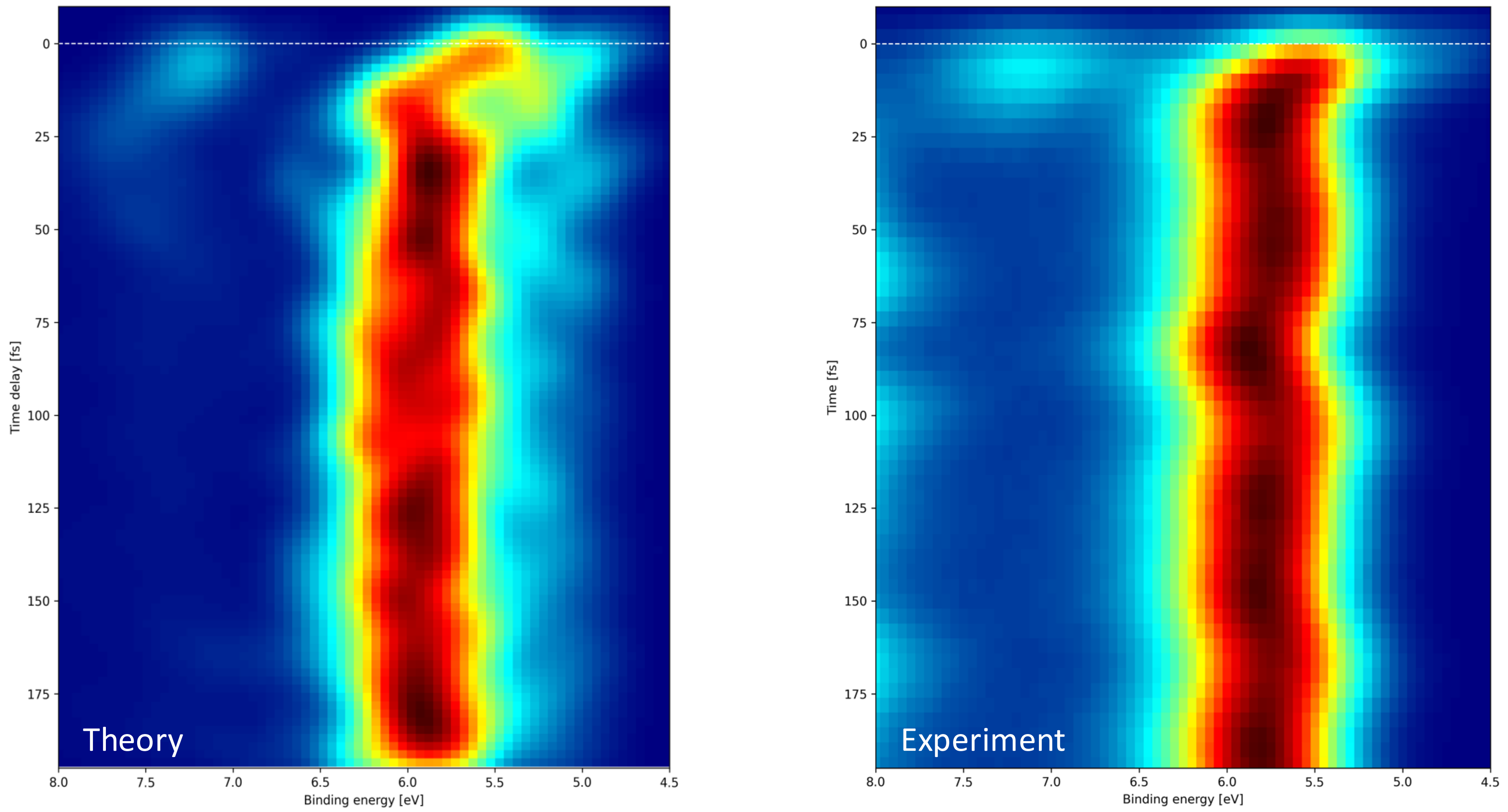
- Generalization to more than two states
- Ultrafast B_{2u} (*ππ**) relaxation into the *nπ** manifold
- Oscillations between A_u (*nπ**) and B_{3u} (*nπ**)
- Decay and oscillation rates remain under debate



4. S. Angelico, E. F. Kjønstad, T. J. Martínez, and H. Koch. 2026, *To be submitted*

Pyrazine

S. Angelico, E. F. Kjønstad, T. J. Martínez, and H. Koch. 2026, *To be submitted*
S. Karashima, A. Humeniuk, and T. Suzuki, 2024, *J. Am. Chem. Soc.*



Outlook

Dynamics in the excited states:

- Powerful new method with accuracy beyond approximate doubles
- But CCSD can struggle even for excited states with single-excitation character
- Perturbative triples: a gold standard for singly-excited-state dynamics (< 0.04 eV)?

Dynamics to the ground state:

- Genuine multireference approaches
- Changing the reference: SF, EA, CVX,[5] *etc.*

5. Rossi, Kjønstad, Angelico, Koch. *J. Phys. Chem. Lett.* 2025, 16, 2, 568–578; see also Kjønstad and Koch, *J. Chem. Phys.* 163, 194104 (2025)

A new tool for the community

Interfaces to nonadiabatic dynamics codes

FMS90

OpenFMS

…?

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